

GridAI

Anti-Herding Compliance for Distributed
Energy Resources

Four agents on Band · Regulated & High-Stakes
Workflows

lablab.ai Band of Agents Hackathon · June 2026

THE PROBLEM

Australia has 15 GWh of home batteries reading the same price signal.

When they all respond at once, they synchronise.

The fleet creates a new evening demand spike instead of smoothing the old one.

471 voltage breaches at the network edge.

This is the herding problem — and it gets worse as VPPs scale.



Current DERMS/VPP platforms (EnergyHub, Kraken, AutoGrid) coordinate DERs — none expose anti-herding assurance.

WHY IT'S HARD

The problem is second-order: it appears precisely when VPP coordination succeeds.

Each aggregator's fleet looks fine in isolation. The failure is at the system level.

WHAT EXISTS	WHAT'S MISSING
DERMS platforms coordinate DERs well	Anti-herding assurance before dispatch
Voltage constraint management is table stakes	Cause-attributed breach compliance
Real-time dispatch of storage fleets	Regulator-ready audit trails

Commercial DERMS expose coordination. None expose anti-herding compliance as a primary artifact.

THE MECHANISM INSIGHT

Desynchronisation is not just a protocol problem.

It depends on fleet-level value heterogeneity. The market design matters as much as the protocol.

NAIVE

1.000

60/60 homes
simultaneous

GRIDAI
HOMOGENEOUS

0.367

22/60 homes
identical fleet

GRIDAI
HETEROGENEOUS

0.167

10/60 homes
realistic fleet

Heterogeneous fleet synchrony is ~55% lower than homogeneous. Market incentives shape the outcome.

THE SOLUTION

GridAI: priority-based coordination protocol with a Band-native compliance layer.

- The Coordinator allocates each battery's dispatch slot from global fleet state, using SOC and owner preference.
- Centralised priority allocation, decentralised-ready. Converges in 1 to 2 rounds.

Battery-herding overvoltage:

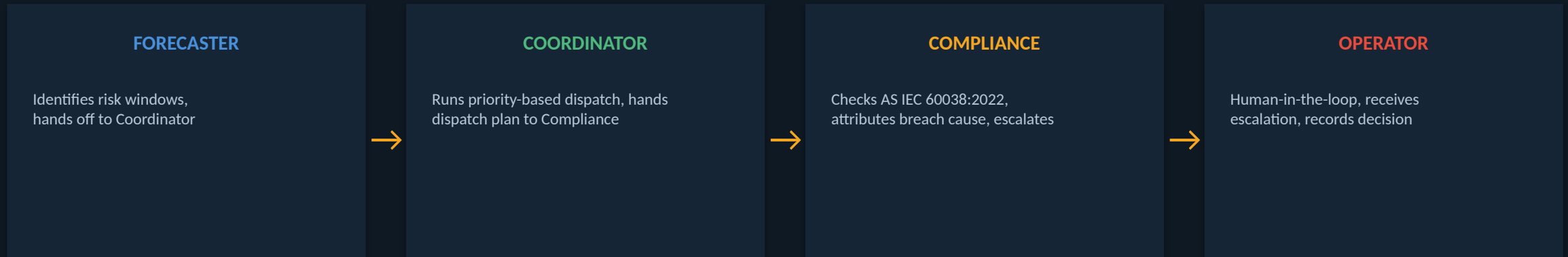
471 → 0

Honest tradeoff: 435 far-feeder undervoltage events (distinct from herding, disclosed not hidden, next tuning target)



THE BAND ARCHITECTURE

Four agents coordinating through Band as the actual collaboration layer.



BAND: shared room · @mention routing · append-only audit log · identity & handoff

Every handoff is traceable. Removing any agent breaks the chain (verified).

Band agents: @s4142972/gridai-forecaster · coordinator · compliance · operator

THE EVIDENCE

Metric	Naive	GridAI
Battery-herding overvoltage	471 steps	0 (eliminated)
Synchrony	1.000	0.167
Max simultaneous discharge	60/60	10/60
Convergence	—	1 to 2 rounds
Peak demand reduction	—	-0.9% (honest)

89 tests passing • including provenance coherence & agent interdependence

AEMO 2012 Victorian data, 17,568 rows. Representative day: 2012-01-24 (highest evening peak, 8,864 MW).

THE COMPLIANCE ARTIFACT

BAND AUDIT TRAIL – NAIVE (COORDINATION & TRANSPORT LAYER)	BAND AUDIT TRAIL – GRIDAI (COORDINATION & TRANSPORT LAYER)
(1) 2026-06-12 05:06:55 forecaster - BAND register register capabilities: ['planning', 'demand_forecast', 'risk_identification']	(1) 2026-06-12 05:06:56 forecaster - BAND register register capabilities: ['planning', 'demand_forecast', 'risk_identification']
(2) 2026-06-12 05:06:55 coordinator - BAND register register capabilities: ['execution', 'priority_coordination', 'dispatch_planning']	(2) 2026-06-12 05:06:56 coordinator - BAND register register capabilities: ['execution', 'priority_coordination', 'dispatch_planning']
(3) 2026-06-12 05:06:55 compliance - BAND register register capabilities: ['review', 'voltage_compliance', 'audit_trail', 'escalation']	(3) 2026-06-12 05:06:56 compliance - BAND register register capabilities: ['review', 'voltage_compliance', 'audit_trail', 'escalation']
(4) 2026-06-12 05:06:55 operator - BAND register register capabilities: ['decision', 'governance', 'human_in_the_loop']	(4) 2026-06-12 05:06:56 operator - BAND register register capabilities: ['decision', 'governance', 'human_in_the_loop']
⚠️ ESCALATE BAND audit · step 7 · 2026-06-12 05:06:56 compliance → operator · handoff:compliance_escalation decision=ESCALATE herding_overvolt=471 reason=battery-herding overvoltage detected: 471 breach events under the naive dispatch Battery-herding overvoltage detected: 471 breach events under the naive dispatch strategy. Naive price-following synchronised the fleet and breached the voltage band. Human operator decision required.	✓ APPROVED BAND audit · step 7 · 2026-06-12 05:06:56 compliance → operator · compliance_approval decision=APPROVED herding_overvolt=0 reason=zero battery-herding overvoltage breaches detected under the priority-based coor Zero battery-herding overvoltage breaches detected under the priority-based coordination protocol. The Coordinator eliminates herding overvoltage while a disclosed residual of 435 far-feeder undervoltage events remains (known tuning target). Dispatch plan approved.

The Band audit trail is the regulated-workflows deliverable.

Every compliance decision is traceable to:

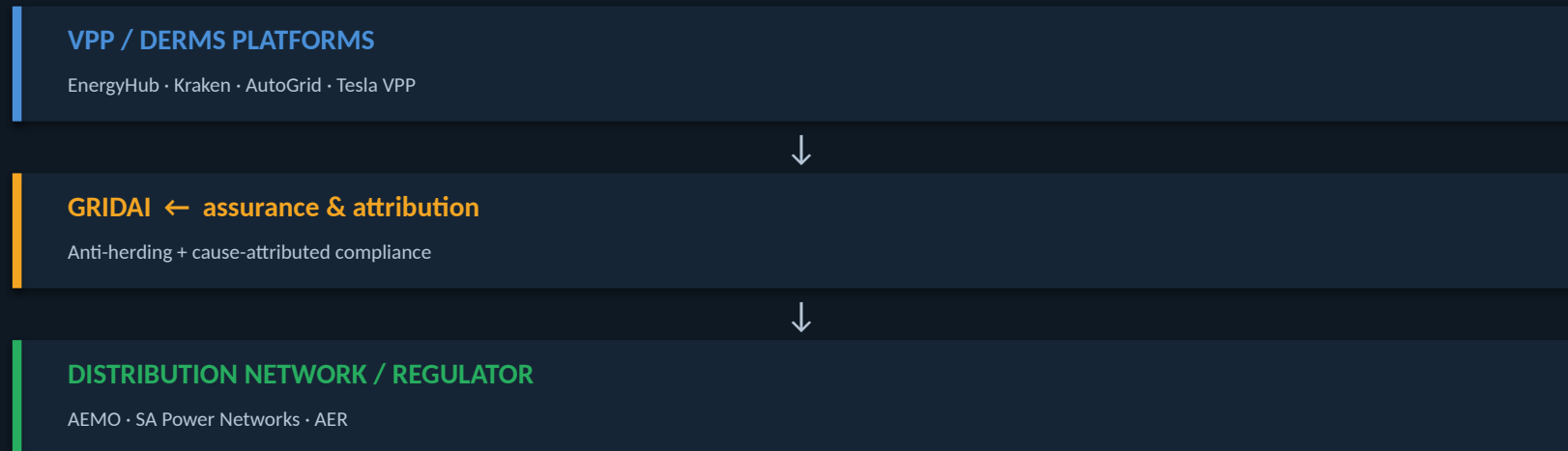
- The agent that made it
- The data it saw
- The moment it happened
- The cause category (pv_export vs battery_herding)

This is what a network operator or regulator can actually use.

WEDGE POSITIONING

GridAI is not a replacement for DERMS or VPP platforms.

It is an assurance and attribution layer that sits between fleets and the network.



NEXT STEPS

- OpenDSS validation
- Multi-aggregator scenarios
- CSIP-AUS live interface
- RAISE Summit Paris, July 2026

Links & Repository

PUBLIC DEMO

<https://dexflex66.github.io/gridai/>

GITHUB

<https://github.com/dexflex66/gridai>

BAND AGENTS

@s4142972/gridai-forecaster · coordinator · compliance · operator

89 tests · 82 original + 7 causal-link & regression additions

Built with: Python · Band SDK · AEMO open data · AS IEC 60038:2022

GridAI heterogeneous 0.167 → GridAI homogeneous 0.367 → Naive 1.000 synchrony baseline